Preface

At its core, this is a book about operations management. If you opened the syllabus of the "core" operations management course in almost any business school in North America, you would find the topics we cover here: processes and flows, capacity and bottlenecks, inventory, variability, waiting lines, quality. These topics form a logical backbone how work gets done, and can be improved systematically, in practice. While the topics are roughly familiar, the way we deliver those topics in this book is dramatically different. Most books on operations management used in business schools march you through these ideas in orderly rows: definition, formula, worked example, and repeat. We've chosen to travel the same territory by a stranger and, we think, more rewarding road.

The road is a conversation. The heart of each chapter is a dialogue between a professor and a handful of students. This format re-enacts what happens in a case-method classroom. The professor is Radu, and he has a maddening, wonderful habit: he refuses to give out answers. Ask him to define a term, and he asks you a question back. Tell him something is obvious, and he'll make you prove it with a piece of chalk. Around him sit a few students who argue, grapple with new ideas and confusions, sometimes get things wrong, and occasionally lose their cool. In their arguing, the ideas of operations management assemble themselves in front of you.

We chose to write it this way for a simple reason: the dialogues replicate how we aim to teach. We believe in this messy approach to coming to understanding. Both of us spend our days in case-method classrooms, and we have come to trust the power of conversation. There is a tempting model of education in which knowledge is a liquid and the student an empty vessel. Teachers pour and students are filled. We believe that the understanding that lasts is the understanding students reach themselves. Ideas they turn over, resisted, and test against their own experience. Ideas they can truly claim to be their own. After all, a definition handed to you is forgotten. A truth you discovered yourself stays. This book is built to let you do the discovering.

There is a second thing that dialogue does, quieter but just as deliberate. As the conversations unfold, the people in them come into focus: the personalities of the participants, their habits of mind. You begin to know who will leap at a problem and who will sit back and doubt it, who trusts a diagram and who trusts a story. You will watch how Radu meets each of them: how he answers the eager student differently from the skeptical one, when he presses and when he waits, how he turns a half-formed objection into the best moment of the class. For an educator, the dialogues serve as a working model of how to navigate a case-method classroom.

A word about pace. The treatement of the material is deliberately slow. Operations management has long carried a reputation as the analytical heavyweight of the business school: the course of formulas, the one non-technical students brace themselves for. Math is there—and we don't avoid it—but it arrives only when the time is right. In particular, we refuse to lead with math. Before any formula, our work is to help you build the intuition and the idea that the cold notation is only trying to capture. A queue is a feeling before it is an equation; a bottleneck is something you've stood in before it is something you compute. Get the intuition first and the math becomes a friend rather than a gatekeeper. We've paced the book to honor that order, which is why we release it gently: one class at a time, every two weeks, in two halves. The first week brings the dialogue, the argument as it happens, figures and all. The second brings the follow-up emails Radu sends to clarify what got left loose, the reflections worth carrying around, and

exercises for those who want to push further. The gap between the two halves of the lesson is not to encourage idleness; it's where understanding settles and becomes yours.

A word, finally, about the authors. Beyond the classroom, we are both researchers in operations management, and we believe deeply in the potential of this subject to make the real world work better. We have built our careers on this belief. But research can make a subject sound bloodless, and operations is anything but. Behind every flow is a worker; behind every wait is a person; behind the cool geometry of an efficient system is a great deal of human care. This book is, in part, our attempt to put that humanity back into view, to show the side of operations that the equations leave out, and, in plain truth, to share our love for a subject we find endlessly, surprisingly alive.

So read it the way the students live it: as a conversation you've been invited into, about a world you thought you already understood. Pour something. Pull up a chair. Read the words out loud, let the characters come alive. Class is about to begin.